

*Bleeding Edge Scientific Foundation
Request for Proposals*

RFP #024B

RAPID DIAGNOSTICS IN RURAL & REMOTE REGIONS

Overview

Issue: *There is a lack of access to emerging healthcare technologies to address rapid diagnostic testing for refugee and displaced populations, particularly in rural or remote areas.*

Motivation: The objective of issuing this award is for increased availability to emerging healthcare technologies to address rapid diagnostic testing for refugee and displaced populations, particularly in rural or remote areas.

Who Are We: The Bleeding Edge Scientific Foundation is a nonprofit aiming to accelerate development and deployment of new technologies to prevent disease in vulnerable populations to promote global health security.

Background

According to the World Health Organization (WHO)¹, 1 in 8 people are on the move globally. Displaced populations, including refugees, have unique and specific health needs that must be considered before, during, and after displacement and migration, and their host community may not be equipped to meet these needs. One report revealed that while migration may not increase the risk of communicable disease transmission in host countries, it does exacerbate migrants' health risks arising from conditions experienced during displacement, e.g., water, sanitation, food, shelter, and essential health services. Nonetheless, refugees and displaced persons experience inequities and disease burdens that cannot be ignored: increasing risk of infectious diseases such as measles, food/water-borne illnesses; gender-based violence, and associated physical and mental health impacts such as PTSD, anxiety, and sleep disorders; and barriers to care due to xenophobia, discrimination, or other obstacles, such as being excluded from national programs.

¹ "Refugee and Migrant Health." n.d. Accessed March 1, 2026.
<https://www.who.int/news-room/fact-sheets/detail/refugee-and-migrant-health>.

A study² on refugee populations in the Southeastern United States, for example, found that of migrants from Africa and Southeast Asia, 76% reported difficulties with paying for food, housing, and healthcare. And there was little change in these numbers a year later, indicating that social determinants of health are consistent and enduring. Refugee health in rural areas, in particular, is complex and multidimensional³, involving challenges such as language barriers, necessitating health literacy programs for culturally competent care delivery. Furthermore, refugees find it difficult to access and navigate the healthcare system in rural areas⁴. They lack funding for basic treatments, and there is a shortage of physicians able and willing to meet the needs of refugee populations worldwide. There is a lack of information sharing regarding the health system in the country of arrival, often leading to inappropriate uses of healthcare systems, e.g., using emergency services instead of a primary care physician.

- Migrant: “any individual who moves across international borders away from his or her country of origin, regardless of legal status or cause.”
- Refugee: “any person who, resulting from a well-founded fear of persecution for reasons of race, religion, nationality, membership of a particular social group or political opinion, is outside the country of his/her nationality and is either unable or too scared to avail himself/herself of the protection of that country.”⁵

Our Foundation aims to address gaps in data on healthcare technologies to address rapid diagnostic testing in rural regions, considering supply chain, manufacturing, point-of-care & point-of-need. Traditional laboratory-based diagnostics are often centralized, infrastructure-dependent, and time-consuming, leading to diagnostic delays that undermine patient care and public health responses. Point-of-care (POC) diagnostics have the potential to provide rapid delivery of actionable results for immediate treatment decisions, reducing the attrition rate from diagnosis to follow-up; and they decentralize testing in high- and low-resource settings.

In rural regions, POC diagnostics bears a unique set of challenges. First, there are persistent trade-offs and deployment barriers. Second, there is often lower analytical sensitivity creating a tension between speed and analytical performance that must be managed through appropriate use-case selection, reflex testing pathways, and quality-assurance programs. Third,

² Feinberg, I., M. H. O'Connor, A. Owen-Smith, and S. R. Dube. 2021. “Public Health Crisis in the Refugee Community: Little Change in Social Determinants of Health Preserve Health Disparities.” *Health Education Research*, February 18, cyab004. <https://doi.org/10.1093/her/cyab004>.

³ Coumans, J. V. F., and S. Wark. 2024. “A Scoping Review on the Barriers to and Facilitators of Health Services Utilisation Related to Refugee Settlement in Regional or Rural Areas of the Host Country.” *BMC Public Health* 24 (January): 199. <https://doi.org/10.1186/s12889-024-17694-9>.

⁴ Spenger, David, Stefan Kordel, and Lukas Schorner. 2025. “Acquiring Health Resources during Settlement in Rural Areas? Refugees’ Experiences of Health Infrastructure and Leisure Practices in Germany.” *Journal of Migration and Health* 13 (December): 100381. <https://doi.org/10.1016/j.jmh.2025.100381>.

⁵ Keidar, Osnat, David S. Srivastava, Emmanouil Pikoulis, and Aristomenis K. Exadaktylos. 2019. “Health of Refugees and Migrants—Where Do We Stand and What Directions Should We Take?” *International Journal of Environmental Research and Public Health* 16 (8): 1319. <https://doi.org/10.3390/ijerph16081319>.

transferring POC approaches to non-communicable disease biomarkers (e.g., cancer protein panels or complex metabolomic signatures) presents further technical hurdles: these analytes often demand higher analytical precision, multi-analyte standardization, and complex pre-analytic handling that currently favor centralized laboratory platforms⁶.

Herein lies an opportunity for institutions to advance diagnostic testing and improve healthcare for refugee and displaced populations. The infectious disease diagnostics market, for example, is estimated to grow 18% over the next five years, from 48.38 billion USD in 2026 to 57.05 billion USD in 2031⁷.

Our Grand Challenge intends to address these gaps and meet the opportunity for rapid diagnostic testing development through the following technical objectives.

Objectives: *A successful application will meet one or more of the following objectives:*

Systems Integration of Diagnostic Data

Focus: improve how the data moves and uses.

- **Objective A:** Develop an API-integrated reporting system that automatically pushes rapid diagnostic results from rural clinics to regional health bodies (like Africa CDC) to trigger the 7-1-7 response framework.
- **Objective B:** Train a response team of local physicians (or community health workers) to use diagnostic data to identify and contain local outbreaks before they reach the scale of an international crisis.

Equity & Sustainability

Focus: ensure emerging technologies are actually affordable and/or locally driven.

- **Objective C:** Design open-source hardware blueprints for diagnostic readers that can be manufactured using locally available materials in rural hubs.
- **Objective D:** Implement a tiered access pricing or manufacturing model where the technology is produced locally within the region to reduce costs by at least 20% compared to imported equivalents.
- **Objective E:** Establish a community-led oversight system in regions with refugees and displaced persons to co-design new diagnostic technologies, ensuring linguistic and cultural literacy.

⁶ Kardjadj, Moustafa. 2025. "Advances in Point-of-Care Infectious Disease Diagnostics: Integration of Technologies, Validation, Artificial Intelligence, and Regulatory Oversight." *Diagnostics* 15 (22): 2845. <https://doi.org/10.3390/diagnostics15222845>.

⁷ Mordor Intelligence. 2026. "Infectious Disease Diagnostics Market Size, Report & Share Analysis 2031." January 15. <https://www.mordorintelligence.com/industry-reports/infectious-disease-diagnostics-market>.

The Bleeding Edge Scientific Foundation (BESF) is a philanthropic organization that funds systems integrators and equity advocates.

Scope

Requests for **funding of up to three years with a maximum budget of 10,000,000 USD** will be considered. Awards will be preferentially issued to applicants who have existing collaborations with refugee or displaced populations from at least one low- or middle-income country (LMIC)⁸, with priorities for displaced populations in countries with high rates of treatable communicable disease.

The Challenge

Objectives: A winning proposal will deliver a validated, ready-to-deploy, rapid diagnostic tool that delivers high precision testing to rural and austere environments. Success will be defined by the development of a diagnostic toolset that functions in low-resource, remote settings and is supported by a sustainable, locally-driven manufacturing and maintenance plan. The tool will provide rapid diagnostics to displaced populations in environments lacking services, equipment, utilities and technologies, commonly found in standard permanent healthcare settings.

Proposal Themes

- **Systems Integration:** Proposals focused on regional frameworks such as the 7-1-7 response framework, connecting rapid diagnostic results to regional health surveillance, such as Africa CDC, via automated digital tools.
- **Engineering & Innovation:** Development of transportable, stand-alone, capable, and ruggedized, hardware that utilizes advanced technologies for diagnostics.
- **Equity & Sustainability:** Proposals establishing local manufacturing hubs, supplementing or building upon open-source hardware to advance local diagnostics, reduce costs and bring equitable healthcare.
- **Operational Research:** Research into overcoming social determinants of health such as language barriers and health literacy, or physical barriers to equitable healthcare such as existing technological mal-alignment, infrastructure improvements and access, within refugee populations.

Process: *To be successful, an application should contain:*

- **Geography:** Lead applicants can be based anywhere in the world, but must include at least one primary partner located in a LMIC where the technology will be deployed.

⁸ As defined by the World Bank Group, see: <https://ourworldindata.org/grapher/world-bank-income-groups>

- **Stakeholders:** Applications must demonstrate a partnership involving at least one academic/research Institution, one non-governmental organization, and one local governmental or local community health organization.

Methodology and Approach

Applicants must provide a phased implementation plan:

1. **Phase I:** Technical validation and prototype ruggedization and austere environmental capability verification.
2. **Phase II:** Regional/cultural adaption (ensure host-region/population compatibility)
3. **Phase III:** Pilot deployment in a remote or refugee-hosting region.

Sustainability and Milestones

Sustainability: Applicants must include a three-year post-funding transition plan, outlining how the technology will be maintained or funded (e.g., through local government adoption or cost-to-user/deployer).

Milestones: Funding will be disbursed in portions based on the achievement of specific milestones, such as a successful field-test with 95% sensitivity or a completion of community-led feedback session with action steps for improvement.

Eligibility: *Applicants must commit to meet the following conditions in order to be considered for funding:*

- All discoveries and data from BESF-funded research must be made public and accessible, as well as compliant with the U.S. Health Insurance Portability and Accountability Act (HIPAA)
- Applications must address at least 20% of the specific objectives outlined in this RFP.
- All applications will be reviewed based on merit, and rewarded based on feasibility.
- Successful applicants will demonstrate existing relationships with NGOs, international organizations, academic institutions, and/or research organizations with relevant research and training missions.
- Organizations must have existing partnerships with an institution(s) in an LMIC.
- Applicants involved in rapid diagnostic development should have reached at least TRL 5 with their technology: [Partial-Scale Prototype & Validation](#)
- The applicants must exhibit a plan that is detailed and demonstrates their commitment to lead and deliver on the final goals and objectives.
- BSEF encourages demonstration of multi-/inter-disciplinary teams that promote a diverse, inclusive, and equitable research environment.

- Researchers are allowed to have co-applicants or collaborators, but ultimately will be required to submit no more than one application as the lead applicant.
- Applicants must agree to the terms and conditions of the funders to be eligible.

Application Requirements: *In addition to eligibility criteria, BESF places specific conditions on applicants to be considered:*

- Must agree with the BESF equitable access policy
- Must agree to follow the binding BESF arbitration process
- Must submit research proposal, equitable partnership statement, and collaboration outline, including existing partnership with an institution in a LMIC
- Must provide rationale for how the proposed intervention accompanies existing case management practices
- Intervention must integrate with both regional surveillance platform as well as an intergovernmental surveillance platform
- Must be able to pass a background check
- Must provide three letters of support showing a similar scope of work
- Leading organization, and no other organization, must submit a single application
- Organizations may not submit more than one application

Exclusions: *We will not consider any submissions that:*

- Do not directly address at least one of the objectives outlined above.
- Do not focus on research or trials intended to address refugee and displaced populations.
- Are submitted by teams that are unable to commit to collaborative efforts with other projects funded by this RFP.
- Are not willing to agree and comply with the BESF data-sharing agreement (see Eligibility).
- Requests a budget exceeding the max of \$10,000,000.

Submission: *All applications must include the following elements:*

- A proposed timeline of the work, to include intermediate milestones reported to the Foundation not less frequently than every six months.
- A budget with breakouts for staff time, travel funding, and equipment.
- A proposed list of core staff and local contacts. An exhaustive list is not necessary.
- A project narrative describing the planned activities and their outcomes.
- Submission of an equitable research collaboration statement as well as sustainability statement outlining how any partnerships will be facilitated.

- Submitters should carefully consider whether there are any significant dual-use concerns to their work and, if so, propose a plan to mitigate these concerns. Submitters should also submit plans, if applicable, to mitigate potential biosafety and bioethics concerns.

References

- Coumans, J. V. F., and S. Wark. 2024. “A Scoping Review on the Barriers to and Facilitators of Health Services Utilisation Related to Refugee Settlement in Regional or Rural Areas of the Host Country.” *BMC Public Health* 24 (January): 199. <https://doi.org/10.1186/s12889-024-17694-9>.
- Feinberg, I., M. H. O’Connor, A. Owen-Smith, and S. R. Dube. 2021. “Public Health Crisis in the Refugee Community: Little Change in Social Determinants of Health Preserve Health Disparities.” *Health Education Research*, February 18, cyab004. <https://doi.org/10.1093/her/cyab004>.
- Kardjadj, Moustafa. 2025. “Advances in Point-of-Care Infectious Disease Diagnostics: Integration of Technologies, Validation, Artificial Intelligence, and Regulatory Oversight.” *Diagnostics* 15 (22): 2845. <https://doi.org/10.3390/diagnostics15222845>.
- Keidar, Osnat, David S. Srivastava, Emmanouil Pikoulis, and Aristomenis K. Exadaktylos. 2019. “Health of Refugees and Migrants—Where Do We Stand and What Directions Should We Take?” *International Journal of Environmental Research and Public Health* 16 (8): 1319. <https://doi.org/10.3390/ijerph16081319>.
- Migration Data Portal. 2025. “Migration and Health.” March 1. <https://www.migrationdataportal.org/themes/migration-and-health>.
- Mordor Intelligence. 2026. “Infectious Disease Diagnostics Market Size, Report & Share Analysis 2031.” January 15. <https://www.mordorintelligence.com/industry-reports/infectious-disease-diagnostics-market>.
- “Refugee and Migrant Health.” n.d. Accessed March 1, 2026. <https://www.who.int/news-room/fact-sheets/detail/refugee-and-migrant-health>.
- Spenger, David, Stefan Kordel, and Lukas Schorner. 2025. “Acquiring Health Resources during Settlement in Rural Areas? Refugees’ Experiences of Health Infrastructure and Leisure Practices in Germany.” *Journal of Migration and Health* 13 (December): 100381. <https://doi.org/10.1016/j.jmh.2025.100381>.